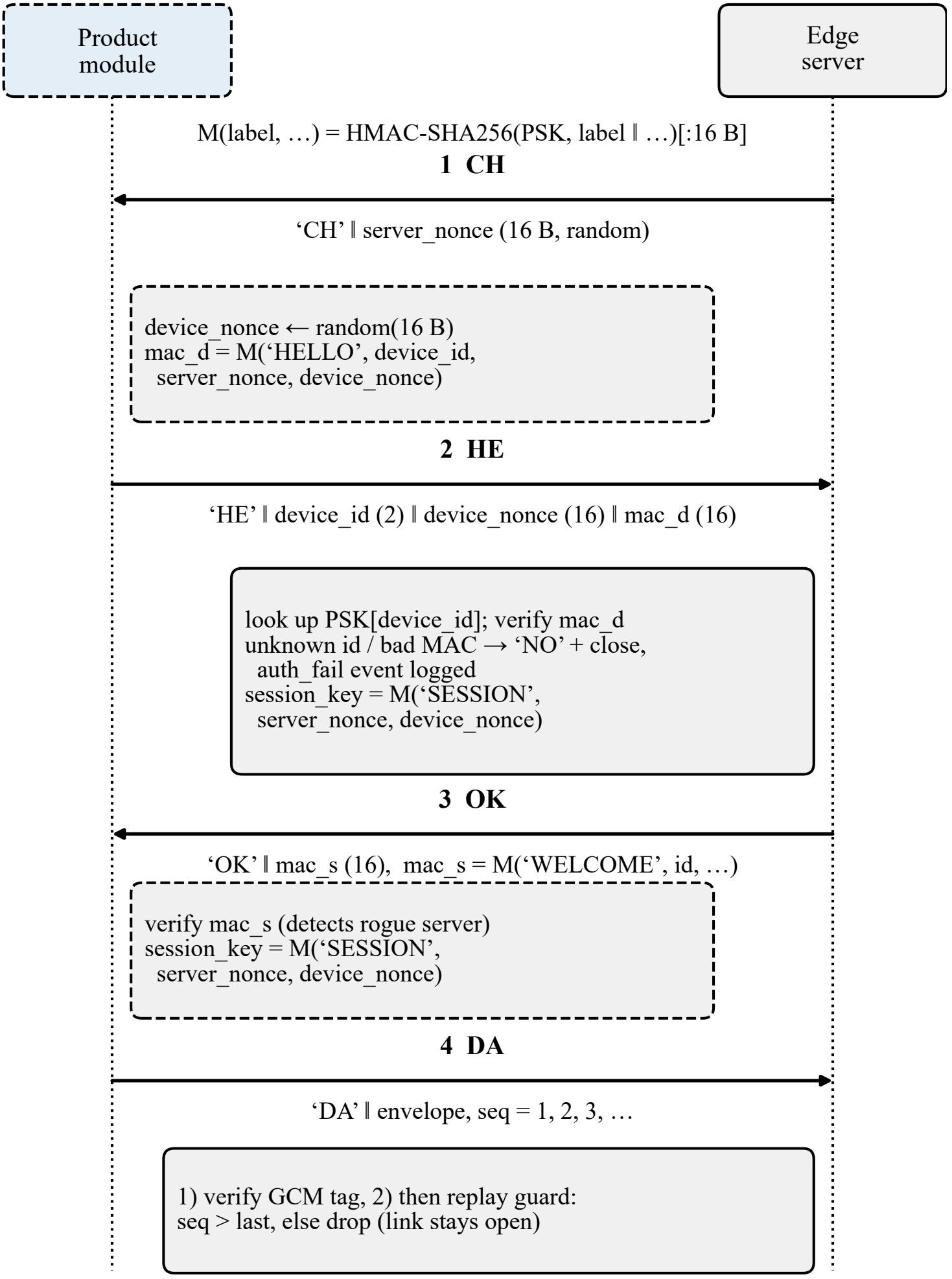
Product module
Edge server
M(label, …) = HMAC-SHA256(PSK, label ‖ …)[:16 B]
1 CH
‘CH’ ‖ server_nonce (16 B, random)
device_nonce ← random(16 B)
mac_d = M(‘HELLO’, device_id, server_nonce, device_nonce)
2 HE
‘HE’ ‖ device_id (2) ‖ device_nonce (16) ‖ mac_d (16)
look up PSK[device_id]; verify mac_d
unknown id / bad MAC → ‘NO’ + close, auth_fail event logged
session_key = M(‘SESSION’, server_nonce, device_nonce)
3 OK
‘OK’ ‖ mac_s (16), mac_s = M(‘WELCOME’, id, …)
verify mac_s (detects rogue server)
session_key = M(‘SESSION’, server_nonce, device_nonce)
4 DA
‘DA’ ‖ envelope, seq = 1, 2, 3, …
1) verify GCM tag, 2) then replay guard:
seq > last, else drop (link stays open)